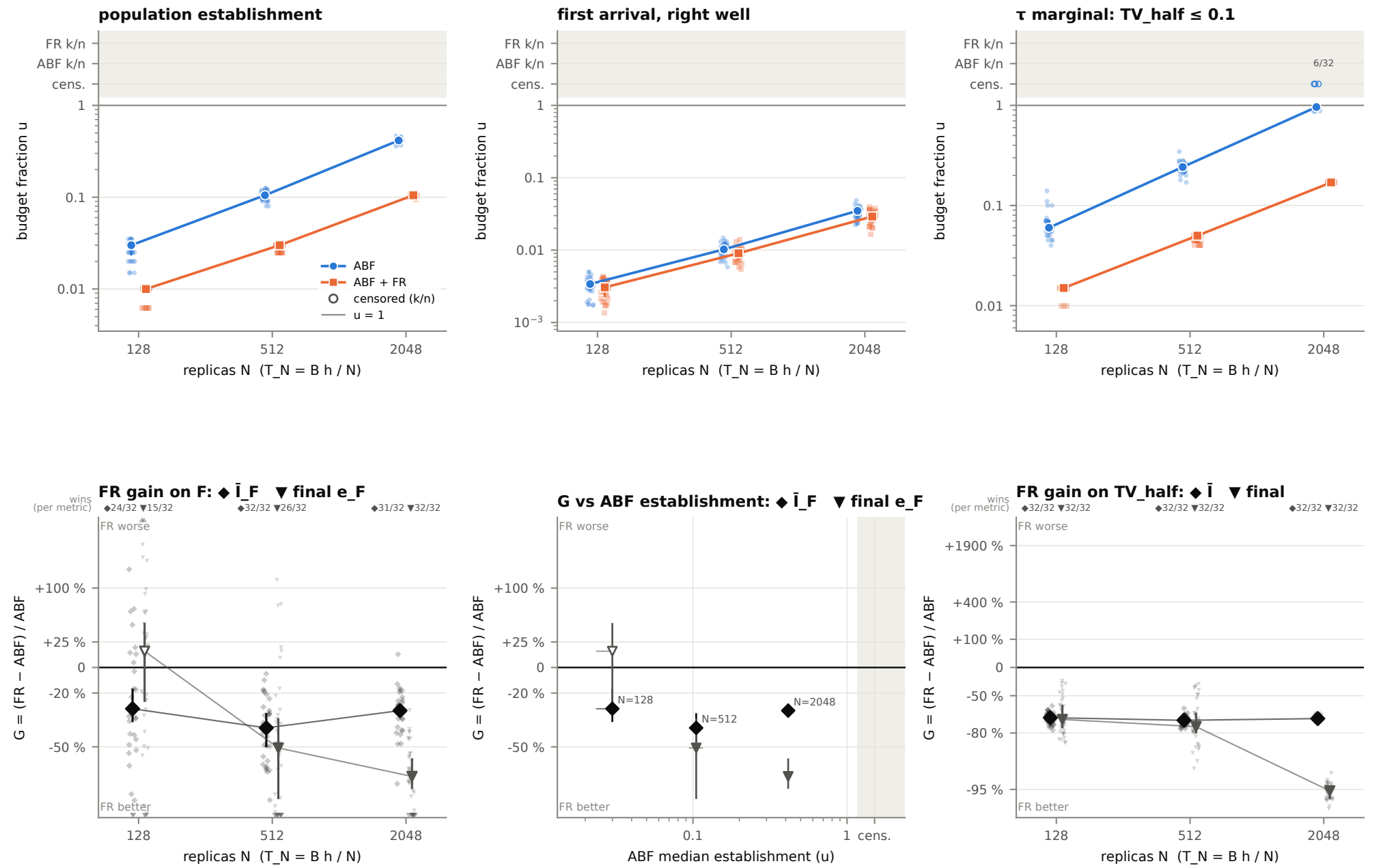


gateway alpha1: population establishment, first arrival and marginal convergence vs N (budget fraction)

Equal total budget per arm per seed: $B = N \times n_steps = 3.2768e+09$ walker-steps at every N ($h = 2.5e-05$, $T_N = B \cdot h / N$); 32 seeds per (N , method). Do FR gains coincide with establishment-limited N ?



gateway alpha1 · S6a_establishment_budget · status: all planned (N , seed, method) complete · FR is not defined at $N = 1$ (ABF only, by design). Establishment = first save after which the instantaneous far-state fraction (right well $x > 0.5$) stays $\geq \frac{1}{2} \times 0.361$. Shaded N [$]$: the instantaneous rule is UNRELIABLE there (ill-defined at $N \leq 2$; an exactly uniform population passes every save of the second half with probability < 0.9 , summary establishment_null), so the same rule on the cumulative visitation is plotted; FR birth-death can push the instantaneous count below binomial scatter, so an FR establishment advantage at those N is not evidence of faster convergence. $\tau(TV_half)$ on the budget axis uses the common budget grid. Top: seed median (large open marker in the 'cens.' band = median censored), IQR bars, faint seeds. Bottom: paired G with bootstrap 95 % CIs (filled = excludes 0; 'wins' = $FR < ABF / n$); scatters: each N 's G against the ABF median time (horizontal bar = ABF IQR; 'cens.' = censored). Source results/mechanism/matched_free_energy/alpha1/analysis/summary.json (generated 2026-10-10T14:19:18Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.